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Student Number



Barker
College

2020

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Advanced

Staff Involved:

- ARP* • LMD • RJW
- RAS • AYG • ESP
- AXD • ALY • JZT

Friday 14th August 2020
8:30 AM

145 copies

General

Instructions:

- Reading time - 10 minutes
- Working time - 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations

Total marks:

100

Section I - 10 marks (pages 2 - 6)

- Attempt Multiple Choice Questions 1 - 10
- Allow about 15 minutes for this section

Section II - 90 marks (pages 7 - 25)

- Attempt Questions 11 - 19
- Show all necessary working
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the derivative of $\ln(\cos x)$?

(A) $-\sec x$

(B) $-\tan x$

(C) $\sec x$

(D) $\tan x$

2. The mean of a set of 10 scores is 14. Another two scores are included and the new mean is 16. What is the mean of the two additional scores?

(A) 4

(B) 16

(C) 18

(D) 26

3. What is the greatest value of the function $f(x) = -2 \cos x + 3$?

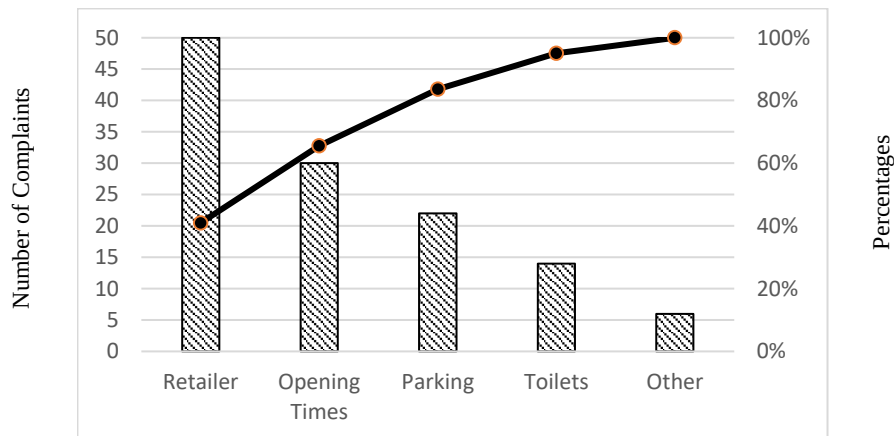
(A) 1

(B) 3

(C) 5

(D) 7

4. Staff at a large department store wanted to determine the key complaints customers had over a period of one month. The following Pareto chart was prepared.



Approximately what percentage of complaints were **Opening Times**?

- (A) 12%
- (B) 25%
- (C) 30%
- (D) 60%
5. The function $f(x) = x^2 - 5x + 1$ is reflected about the y - axis. What is the equation of the new function?

- (A) $f(x) = x^2 + 5x + 1$
- (B) $f(x) = x^2 + 5x - 1$
- (C) $f(x) = -x^2 + 5x + 1$
- (D) $f(x) = -x^2 + 5x - 1$

6. Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = xe^x$ between $x = 1$ and $x = 3$?

(A) $\frac{1}{4}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

(B) $\frac{1}{4}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

(C) $\frac{1}{2}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

(D) $\frac{1}{2}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

7. In a Year 12 drama class of 26 students, all students do either music or art or both. 11 do music. 5 do both music and art. What is the probability that a randomly chosen student does music and art, given they do art?

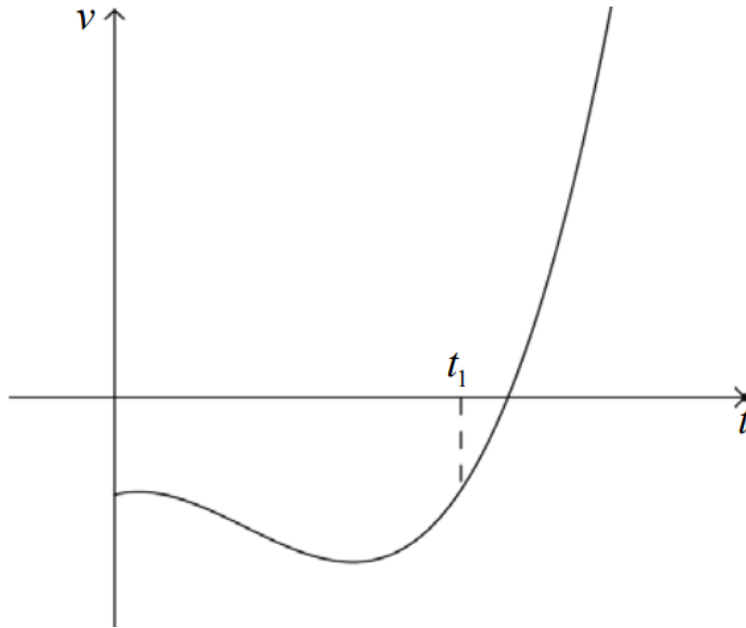
(A) $\frac{5}{26}$

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) $\frac{5}{11}$

8. The graph below shows the velocity v of a particle moving in a straight line as a function of time t . The positive direction of the motion is to the right.



Which statement describes the motion of the particle when $t = t_1$?

- (A) The velocity is positive, and acceleration is positive
- (B) The velocity is negative, and the acceleration is positive
- (C) The velocity is positive, and the particle is to the right of its initial position
- (D) The velocity is negative, and the particle is to the right of its initial position

9. The first three terms of a geometric series are $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots$.

If the series has a limiting sum, then which of the below is true?

(A) $x < -1$ or $x > 1$

(B) $-1 < x < 1$

(C) $|x| < 1$

(D) None of the above

10. The function $y = f(x)$ has a turning point at (6,2). It is transformed to $y = 3f(-2x + 4) + 2$.
Where is the new turning point?

(A) (-1,8)

(B) (-7,8)

(C) (-10,8)

(D) (-16,8)

End of Section I

Section II

90 marks

Attempt Questions 11 - 19

Allow about 2 hours and 45 minutes for this section

In Questions 11–19, your responses should include relevant mathematical reasoning.

Question 11 (10 marks)

(a) Differentiate the following functions.

(α) $(3x + 4)^5$. **1**

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(β) $\frac{\tan x}{x}$. **1**

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(b) (i) Differentiate $y = xe^{3x}$. **1**

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(ii) Hence find the exact value of $\int_0^2 e^{3x}(3+9x) \, dx$. **2**

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(c) A balloon drifts 100 km from point A to point B on a bearing of 28° . At point B the balloon changes direction and drifts 50 km to point C on a bearing of 114° .

(i) Draw a diagram showing the above information.

1



(ii) Show that the distance from point A to point C is 115 km to the nearest kilometre.

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(iii) Find the bearing of point A from point C . Give your answer to the nearest degree.

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Question 12 (10 marks)

Evaluate the following integrals:

(a) $\int 3^x dx.$ 1

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(b) $\int \frac{x}{x^2 + 3} dx.$ 2

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(c) $\int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx.$ 2

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(d) During a storm, water flows into a 7000 litre tank at a rate of $\frac{dv}{dt}$ litres per minute, where $\frac{dv}{dt} = 120 + 26t - t^2$ and t is the time in minutes since the storm began.

(i) At what times is the tank filling at twice the initial rate? 2

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(ii) Initially the tank contains 1500 litres of water. When the storm finishes, 30 minutes after it began, the tank is overflowing. 3

How many litres of water have been lost?

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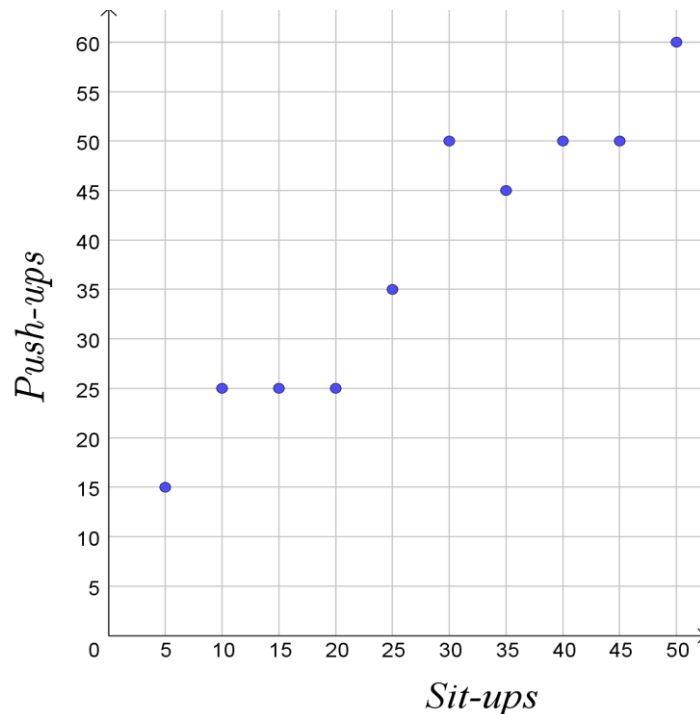
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Question 13 (10 marks)

- (a) The scatterplot below shows the number of Push-ups (P) and the number of Sit-ups (S) performed by 10 students during a fitness test.



- (i) Calculate the value of the correlation coefficient correct to 2 decimal places. 2
 Comment on the strength and direction of correlation.

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- (ii) Determine the equation of the least-squares regression line for this data 2
 in terms of Push-ups (P) and Sit-ups (S).

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- (iii) Predict the number of sit-ups a person can do if they can do 100 push-ups. 1

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(b) An infinite geometric series has a first term of 8 and a limiting sum of 12.

(i) Calculate the common ratio.

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(ii) Hence, calculate the sum of the first three terms.

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(c) In an arithmetic series, the sum of the first 16 terms is 288 and the sixth term is 8.

Find the first term and the common difference.

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Question 14 (10 marks)

(a) A table of future value interest factors for an annuity of \$1 is shown below.

Period	Interest rate per period				
	2.5%	3%	3.5%	4%	4.5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0250	2.0300	2.0350	2.0400	2.0450
3	3.0756	3.0909	3.1062	3.1216	3.1370
4	4.1525	4.1836	4.2149	4.2465	4.2782
5	5.2563	5.3091	5.3625	5.4163	5.4707
6	6.3877	6.4684	6.5502	6.6330	6.7169

- (i) Jenna is saving for a holiday by contributing \$500 every six months into an annuity that pays interest at the rate of 9% p.a., compounded every six months. Use the table above to determine how much she will have in 3 years time. 1

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- (ii) Toby is saving for the same trip but needs \$5000 in total. How much more than Jenna does he need to contribute every six months, using the same interest rate, if he wishes to have enough money in 3 years time? 2

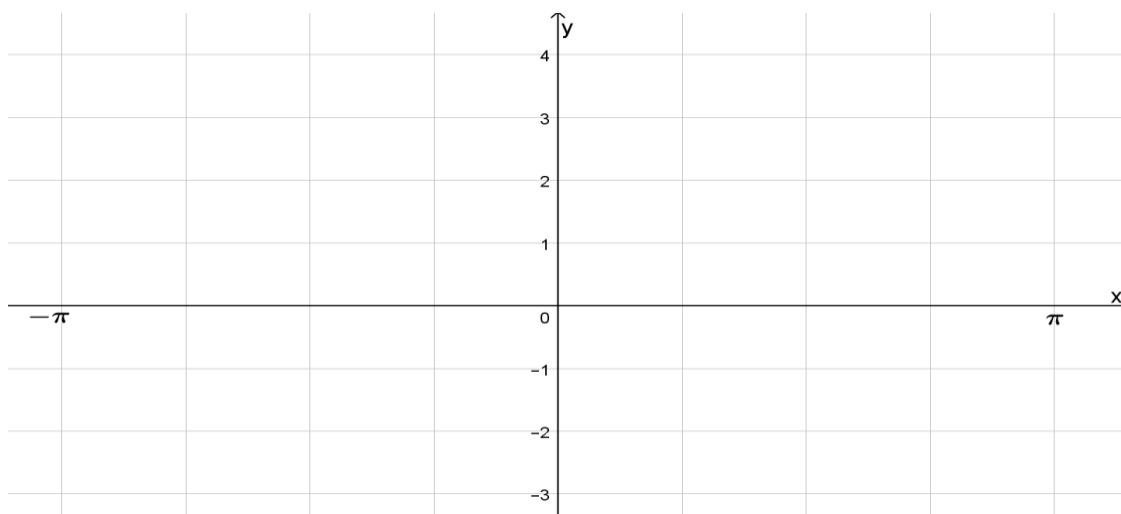
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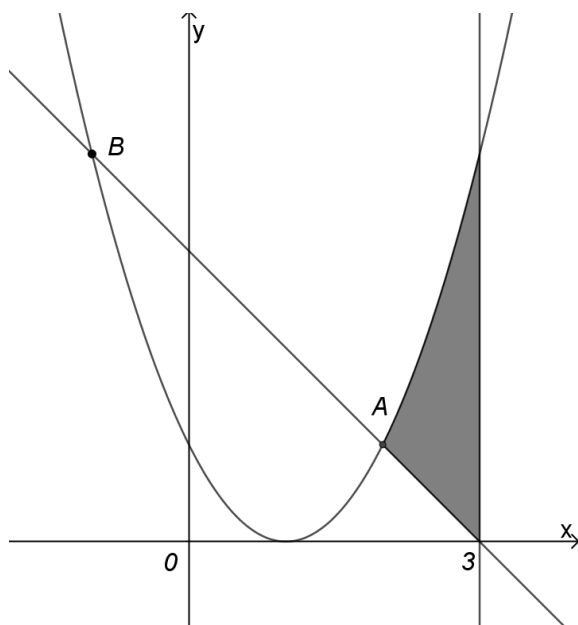
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- (b) On the axes below sketch the function $f(x) = 2 \sin 2x + 1$ in the domain $-\pi \leq x \leq \pi$. (You do not need to show x -intercepts). 3



- (c) The diagram below shows the curve $y = (x - 1)^2$ intersecting with the line $x + y = 3$ at the points A and B .



- (i) Verify that the coordinates of A are $(2, 1)$. **1**

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- (ii) Hence find the shaded area enclosed by the curve $y = (x - 1)^2$ **3**
and the lines $x + y = 3$ and $x = 3$.

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Question 15 (10 marks)

(a) Let $f(x) = 3x^4 - 8x^3 + 6$.

- (i)

Find the coordinates of the stationary points of $y = f(x)$,
and determine their nature.

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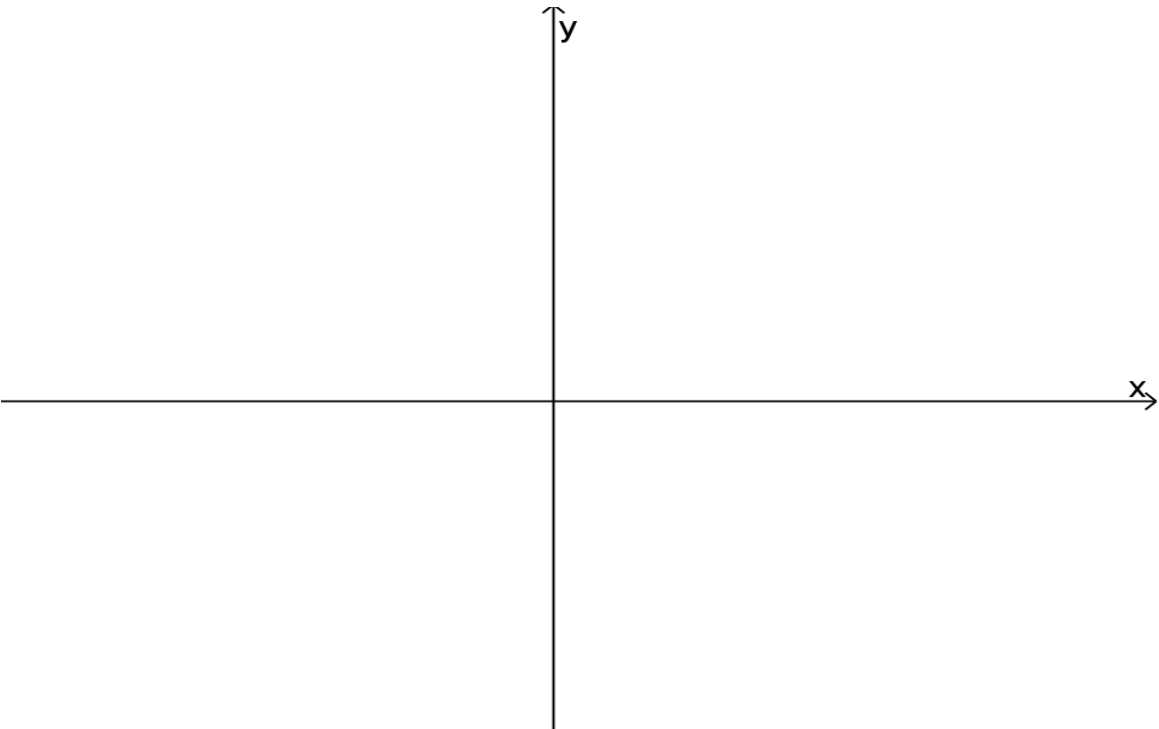
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- (ii)

Hence, sketch the graph of $y = f(x)$ showing all stationary points.

2



(b) For the probability distribution below $E(X) = 3.34$.

x	1	3	a	6
$p(x)$	0.4	0.12	0.3	b

(i) Evaluate a and b .

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(ii) Evaluate the Variance.

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Question 16 (10 marks)

- (a) Due to overfishing in a particular bay, the number (N) of a particular species of fish is dropping exponentially according to the formula $\frac{dN}{dt} = -kN$, where time (t) is measured in years after 1930. It is known that in 1930 there were 25,000 fish of this species and by 2020 there were only 2000 fish.

- (i) Show that $N = Ae^{-kt}$, where A and k are constants, satisfies the equation $\frac{dN}{dt} = -kN$. **1**

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- (ii) Find the value of A and show that $k \div 0.028$. **2**

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- (iii) This species of fish will be declared extinct in this bay when the number drops below 500 fish. In which year will this occur? **2**

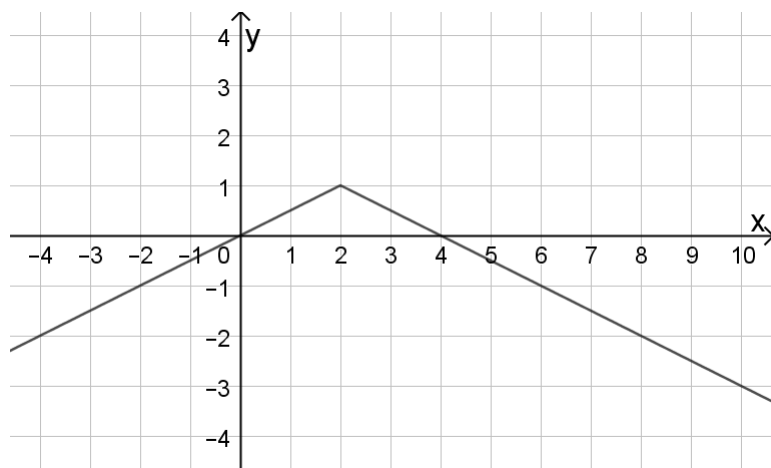
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Question 17 (10 marks)

- (a) The function $f(x) = |x|$ is transformed and the equation of the new function is of the form $y = kf(x + b) + c$ where k , b and c are constants.

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The graph of the new function is shown below.



What are the values of k , b and c ?

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- (b) (i) When Josephine started a new job, \$450 was deposited into her superannuation fund at the beginning of each month. The money was invested at 0.5% per month, compounded monthly.
Let P be the value of the investment after 300 months, when Josephine retires.

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Show that $P = \$313\,406.52$.

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Question 17 (b) continues on the following page

- (ii) After retiring, Josephine withdraws \$2500 from the account at the end of each month, without making any further deposits. The account continues to earn interest at 0.5% per month.

Let A_n be the amount left in the account n months after Josephine's retirement.

- (α) Show that $A_n = (P - 500\,000) \times 1.005^n + 500\,000$ where P was defined in (i).

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- (β) For how many months after retirement will there be money left in the account?

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Question 18 (10 marks)

(a) A particle is moving in a straight line with velocity $v = 3e^t + 6e^{-t}$.
It begins its motion at the Origin O , t is in seconds and v is metres per second.

(i) Find the displacement function, x , of the particle, at the time t seconds. 2

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(ii) Find the exact time when the particle is at $x = 10$ metres. 3

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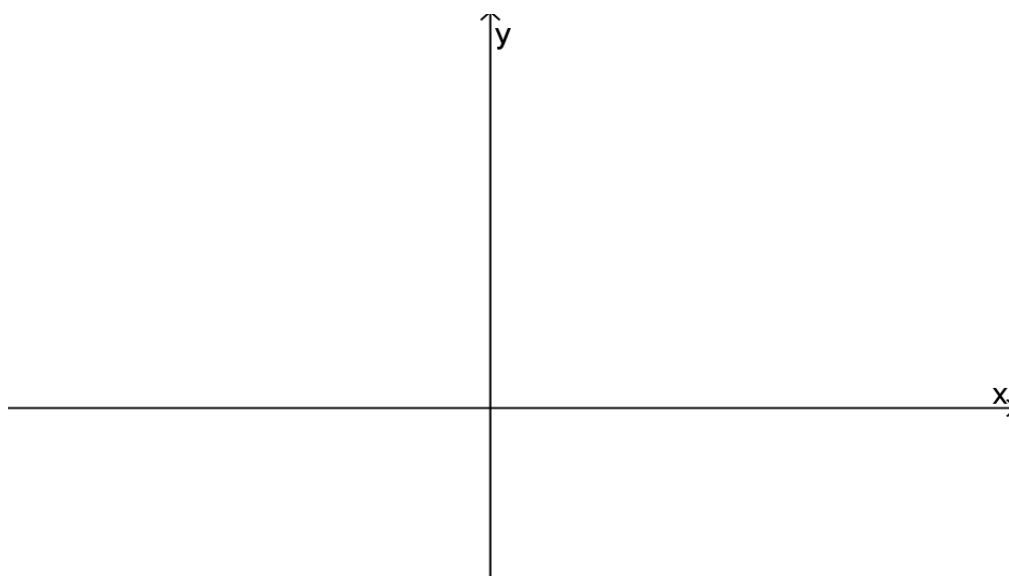
(b) The function $y = e^x$ is reflected about the y –axis and moved up by 1 unit to give the function $f(x)$.

(i) Write down the equation of the function $f(x)$. 1

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(ii) Sketch $y = f(x)$ on the number plane below showing all important details. 1



(iii) Consider the function $g(x) = \ln x$. 3

Determine the composite function $f(g(x))$ indicating its domain and range.

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Question 19 (10 marks)

(a) A student tosses two regular six-sided dice and writes down the two numbers showing on the uppermost faces.

(i) Find the probability that their sum is greater than 7 and an even number. 2

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(ii) Find the probability that their sum is greater than 7 and an even number, 2
given that at least one of the numbers is even.

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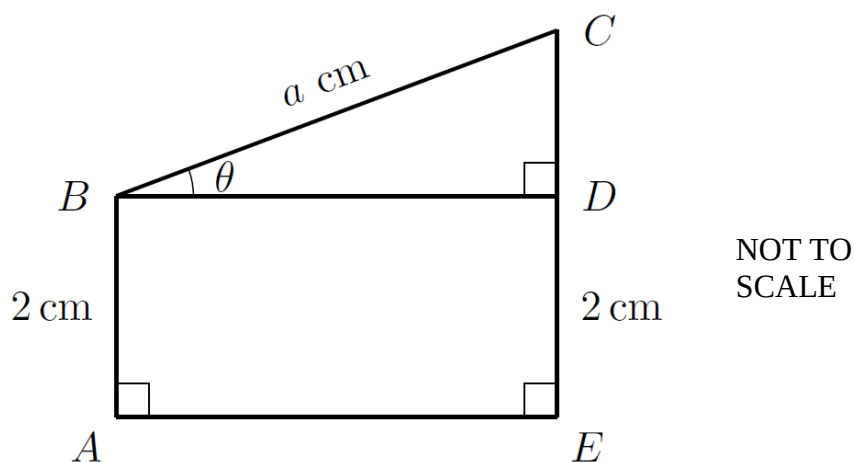
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- (b) The figure shown represents a wire frame made up of 6 parts; AB, BC, CD, ED, AE and BD .
 $AB = ED = 2 \text{ cm}$ and $BC = a \text{ cm}$ where $a > 0$.
 $\angle CBD = \theta$, where $0 < \theta < \frac{\pi}{2}$



- (i) Find BD and CD in terms of a and θ . 1

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- (ii) Find the length L of the wire frame in terms of a and θ . 1

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Question 19 (b) continues on the following page

- (iii) Given that $a = 3\sqrt{5}$ find the exact maximum possible length of wire in the frame. 4

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END OF PAPER

MULTIPLE CHOICE

1. $y = \ln(\cos x) \quad y' = \frac{-\sin x}{\cos x}$
 $= -\tan x$ (B)

2. $14 \times 10 = 140$
 $16 \times 12 = 192$
 $\text{DIFF} = 52 \quad \bar{x} = 26$ (D)

3. WHEN $x = -1$
 $f(-1) = -2x - 1 + 3$
 $= 5$ (C)

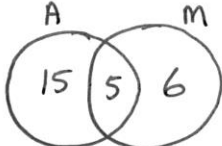
4. APPROX $65\% - 40\% = 25\%$ (B)

5. REFLECT ABOUT y :
 $f(-x) = (-x)^2 - 5(-x) + 1$
 $= x^2 + 5x + 1$ (A)

6. $h = \frac{3-1}{4} = \frac{2}{4} = \frac{1}{2}$

x	1	1.5	2	2.5	3
y_n	e^1	$1.5e^{1.5}$	$2e^2$	$2.5e^{2.5}$	$3e^3$
	y_0	y_1	y_2	y_3	y_4

$$A \approx \frac{1}{2} \left[e^1 + 3e^3 + 2(1.5e^{1.5} + 2e^2 + 2.5e^{2.5}) \right]$$
$$= \frac{1}{4} (e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$$
 (B)

7. 
 $\therefore \frac{5}{20} = \frac{1}{4}$ (B)

8. $\cdot V$ IS -VE ... BELOW x -AXIS
 $\cdot A$ IS +VE ... +VE GRADIENT (B)

9. $r = \frac{1}{x^2} \div \frac{1}{x} = \frac{1}{x}$
 $\therefore -1 < \frac{1}{x} < 1$
WHICH MEANS $x < -1$ or $x > 1$ (A)

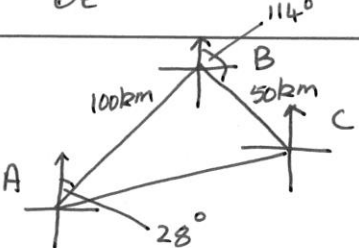
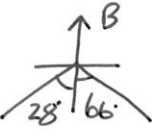
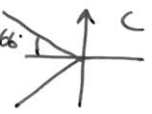
10. $y = 3f(-2[x-2]) + 2$
 $\begin{array}{cccc} \uparrow & \uparrow & \uparrow & \uparrow \\ \text{VERT.} & \text{HORIZ.} & \text{RIGHT} & \text{UP} \\ \text{DILATE} & \text{DILATE} & 2 & 2 \\ \text{(3)} & \text{(1)} & \text{(2)} & \text{(4)} \end{array}$
 x COORD: $6 \times -1/2 = -3$ (1)
 $-3 + 2 = -1$ (2)
 y COORD: $2 \times 3 = 6$ (3)
 $6 + 2 = 8$ (4)
 $\therefore (-1, 8)$ (A)

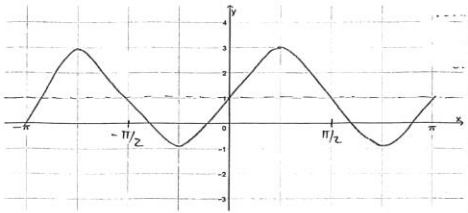
QUESTION 11.

a. $y = (3x+4)^5$
 $y' = 5 \cdot 3 \cdot (3x+4)^4 = 15(3x+4)^4$

b. $y = \frac{\tan x}{x} \quad u = \tan x \quad v = x$
 $u' = \sec^2 x \quad v' = 1$
 $y' = \frac{x \sec^2 x - \tan x}{x^2}$

b. i. $y = xe^{3x} \quad u = x \quad v = e^{3x}$
 $u' = 1 \quad v' = 3e^{3x}$
 $y' = 3xe^{3x} + e^{3x}$
 $= e^{3x}(3x+1)$

ii.	$\int_0^2 e^{3x} (3+9x) dx$ $= \int_0^2 3e^{3x} (3x+1) dx$ $= 3 \int_0^2 e^x (3x+1) dx$ $= 3 \left[x e^{3x} \right]_0^2$ $= 3 (2e^6 - 0)$ $= 6e^6$	QUESTION 12	
		a.	$\int 3^x dx = \frac{1}{\ln 3} \cdot 3^x + C.$
		b.	$\int \frac{x}{x^2+3} dx = \frac{1}{2} \int \frac{2x}{x^2+3} dx$ $= \frac{1}{2} \ln(x^2+3) + C$
		c.	$\int_0^{\pi/2} \sin \frac{x}{2} dx = \left[-2 \cos \frac{x}{2} \right]_0^{\pi/2}$ $= (-2 \cos \pi/4) - (-2 \cos 0)$ $= -\frac{2}{\sqrt{2}} + 2$ RATIONALISE ETC.. $= 2 - \sqrt{2} \text{ OR } 0.587..$
c.i.			
ii.	 $x^2 = 100^2 + 50^2 - 2 \cdot 100 \cdot 50 \cdot \cos 94^\circ$ $= 114.88...$ $= 115 \text{ km.}$	di.	INITIAL RATE: $t=0 \quad \frac{dV}{dt} = 120 + 0 + 0$ $= 120 \text{ L/MIN}$ $\therefore 120 \times 2 = 240$ $\therefore 240 = 120 + 26t - t^2$ $t^2 - 26t + 120 = 0$ $(t-20)(t-6) = 0$ $\therefore t = 20 \text{ MINS, } 6 \text{ MINS.}$
iii.	 $\angle BCA = \frac{\sin 94^\circ}{115} = \frac{\sin x}{100}$ $\sin x = \frac{100 \sin 94^\circ}{115}$ $= 60.1^\circ$ $\text{BEARING} = 360^\circ - 66^\circ - 60.1^\circ$ $= 233^\circ 50' = 234^\circ$	ii.	$V = 120t + 13t^2 - \frac{t^3}{3} + C$ $t=0 \quad 1500 = 0 + 0 - 0 + C$ $V=1500 \quad \therefore C=0$ $\text{EQ'N: } V = 120t + 13t^2 - \frac{t^3}{3} + 1500$ PTO →

$t=30$ $V = 3600 + 11700 - 9000 + 1500$ $= 7800$ $7000 - 7800 = -800L$ ie: 800L OVERFLOWED.	c. $T_6 = 8 = a + 5d$ — ① $S_{16} = 288 = \frac{16}{2} [2a + 15d]$ $288 = 8(2a + 15d)$ $288 = 16a + 120d$ — ②
<u>QUESTION 13.</u> ai. PUT IN CALC... $r = 0.96$ STRONG, POSITIVE CORRELATION ii. PUT IN CALC... $A = 12$ $B = 0.945...$ $P = 0.95S + 12$ iii. $100 = 0.95 \times S + 12$ $\therefore S = 92.6$ $= 93 \text{ SITUPS}$	$① \times 16: 128 = 16a + 80d$ ③ $② - ③: 160 = 40d$ $\therefore d = 4$ SUB INTO ①: $8 = a + 20$ $\therefore a = -12$
bi $a=8$ $12 = \frac{8}{1-r}$ $S_{\infty} = 12$ $12(1-r) = 8$ $12 - 12r = 8$ $-12r = -4$ $r = \frac{1}{3}$ ii. $S_3 = ?$ $a=8$ $S_3 = \frac{8(\frac{1}{3}^3 - 1)}{\frac{1}{3} - 1}$ $r = \frac{1}{3}$ $= \frac{104}{9}$ (U)	<u>QUESTION 14.</u> ai. $9\% \div 2 = 4.5\%$ $3 \times 2 = 6 \text{ PERIODS}$ $\therefore \$500 \times 6.7169$ $= \$3358.45$ ii. $\$5000 \div 6.7169 = \744.39 $\$744.39 - \$500 = \$244.39$
	b. 

ci. SOLVE SIMULTANEOUSLY:

$$(x-1)^2 = 3-x$$

$$x^2 - 2x + 1 = 3 - x$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$\therefore x = 2, -1$$

SUB. IN $x=2$ TO $x+y=3$

$$\therefore y = 1$$

$\therefore (2, 1)$ LIES ON BOTH.

ii. AREA = TOP - BOTTOM

$$A = \int_2^3 (x-1)^2 - (3-x) dx$$

$$= \int_2^3 x^2 - x - 2 dx$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_2^3$$

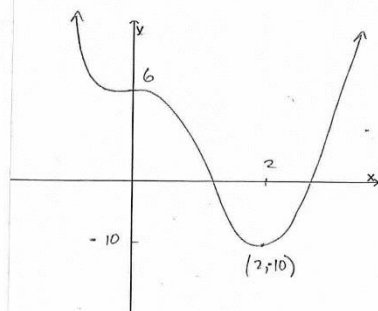
$$= (9 - 9/2 - 6) - (8/3 - 2 - 4)$$

$$= 11/6 \text{ UNITS}^2.$$

$$f''(2) = 144 - 96 \\ = 48 \dots \text{+VE } \cup$$

$\therefore \text{MIN @ } (2, -10)$

HORIZ POI @ $(0, 6)$



bi.

x	1	3	a	6	SUM
p(x)	0.4	0.12	0.3	b	1

$$x p(x) \quad 0.4 \quad 0.36 \quad 0.3a \quad 1.08 \quad 3.34 \leftarrow \mu \text{ or } E(x)$$

$$x^2 p(x) \quad 0.4 \quad 1.08 \quad 7.5 \quad 6.48 \quad 15.46 \leftarrow E(x^2)$$

$$b = 1 - (0.4 + 0.12 + 0.3) \\ = 0.18$$

$$0.3a = 3.34 - (0.4 + 0.36 + 1.08)$$

$$0.3a = 1.5$$

$$\therefore a = 5$$

$$\text{ii. Var}(x) = E(x^2) - \mu^2 \\ = 15.46 - 3.34^2 \\ = 4.3044$$

QUESTION 15.

ai. $f'(x) = 12x^3 - 24x^2$

STAT: $0 = 12x^3 - 24x^2$

PT $0 = 12x^2(x-2)$

$$\therefore x = 0, 2$$

$$f''(x) = 36x^2 - 48x$$

$$f''(0) = 0 \quad \text{TEST IT!}$$

x	-1	0	1
f''(x)	84	0	-12

HORIZ.
 $\therefore \text{POI}$

QUESTION 16.

$$\begin{aligned} \text{i. } \frac{dN}{dt} &= -k \cdot A e^{-kt} \\ &= -kN \text{ as } N = A e^{-kt}. \end{aligned}$$

$$\text{ii. } t=0 \quad 25000 = A e^0 \\ \therefore A = 25000$$

$$2000 = 25000 e^{-90k}$$

$$0.08 = e^{-90k}$$

$$\frac{\ln(0.08)}{\ln(e)} = -90k$$

$$-2.52 = -90k$$

$$\therefore k = 0.0280 \dots \\ = 0.028$$

$$\text{iii. } 500 = 25000 e^{-0.028t}$$

$$0.02 = e^{-0.028t}$$

$$\frac{\ln(0.02)}{\ln(e)} = -0.028t$$

$$-3.91 = -0.028t$$

$$139.71 = t$$

$$\therefore t = 140 \text{ YEARS}$$

$$\therefore \text{DATE} = 2070 \text{ AD.}$$

b. OLD SCHOOL!

$$\frac{40^\circ}{360^\circ} \times 2 \times \pi \times 6 \text{ cm}$$

$$= \frac{1}{9} \times 12\pi$$

$$= \frac{4\pi}{3} \text{ cm. (OR USE } l=r\theta)$$

$$\text{c. } 2\cos^2 x - \cos x - 1 = 0$$

$$\text{Let } u = \cos x$$

$$2u^2 - u - 1 = 0$$

$$(2u+1)(u-1) = 0$$

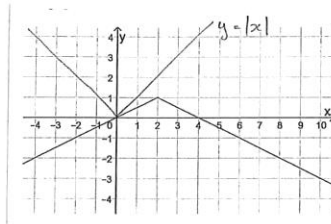
$$\therefore u = -1/2, 1$$

$$\cos x = -1/2 \quad (\text{USE ASTC}) \\ \therefore x = 2\pi/3, 4\pi/3$$

$$\cos x = 1 \quad (\text{USE GRAPH}) \\ \therefore x = 0, 2\pi$$

QUESTION 17.

a.



• FLIPPED ABOUT $x \neq$

"WIDENED" ... $k = -1/2$

• MOVED UP 1 ... $c = 1$

• MOVED RIGHT 2 ... $b = -2$

PTD →

bi. ANNUITIES Q'N...

$$A_1 = 450 (1.005)^{300}$$

$$A_2 = 450 (1.005)^{299}$$

...

$$A_{300} = 450 (1.005)^1$$

$$\therefore A_{\text{TOTAL}} = 450 (1.005^{300} + 1.005^{299} + \dots + 1.005^1)$$

GP! $\swarrow$

$$S_{300} = \frac{1.005 (1.005^{300} - 1)}{0.005}$$

$$a = 1.005 \quad \nearrow$$

$$r = 1.005$$

$$n = 300$$

$$\therefore A_{\text{TOTAL}} = 450 \times 696.45 \dots$$

$$= \$313\,406.52$$

$$A_n = P(1.005)^n - \frac{2500(1.005^n - 1)}{0.005}$$

$$= P(1.005)^n - 500\,000(1.005^n - 1)$$

FACTOR -1SE $\left[\begin{aligned} &= P(1.005)^n - 500\,000(1.005^n) + 500\,000 \\ &\rightarrow = (P - 500\,000) \times 1.005^n + 500\,000 \end{aligned} \right.$

β. $D = (P - 500\,000) \times 1.005^n + 500\,000$

$$- \frac{500\,000}{1.005^n} = 1.005^n$$

$$\frac{P - 500\,000}{1.005^n} = 1.005^n$$

$$P - 500\,000 = 500\,000 \times 1.005^n$$

(= 313 406.52 FROM i.)

$$2.6796 \dots = 1.005^n$$

$$\frac{\ln(2.6796)}{\ln(1.005)} = n$$

$$\therefore n = 197.62 \text{ MONTHS}$$

$$= 197 \text{ or } 198.$$

ii. $A_1 = P(1.005) - 2500$

α. $A_2 = A_1(1.005) - 2500$

$$= \sum P(1.005) - 2500 \times 1.005 - 2500$$

$$= P(1.005)^2 - 2500(1.005 + 1)$$

$$A_3 = P(1.005)^3 - 2500(1.005^2 + 1.005 + 1)$$

... SHOW MORE WORKING THAN ABOVE. I AM LIMITED BY SPACE!

$$A_n = P(1.005)^n - 2500(1.005^{n-1} + \dots + 1)$$

GP! $\swarrow$

$$a = 1$$

$$r = 1.005$$

$$n = n$$

$$S_n = \frac{1(1.005^n - 1)}{0.005}$$

QUESTION 18.

$$\int 3e^t + 6e^{-t}$$

$$= 3e^t - 6e^{-t} + C$$

$$t=0 : 0 = 3 - 6 + C$$

$$x=0 : 0 = -3 + C$$

$$\therefore C = 3$$

$$\therefore x = 3e^t - 6e^{-t} + 3$$

ii. $10 = 3e^t - 6e^{-t} + 3$

$$7 = 3e^t - 6e^{-t}$$

$$7 = 3e^t - \frac{6}{e^t} \quad \left. \vphantom{7 = 3e^t - \frac{6}{e^t}} \right\} \times e^t$$

$$7e^t = 3e^{2t} - 6$$

$$0 = 3e^{2t} - 7e^t - 6 \quad \text{PTD} \nearrow$$

$$\text{Let } u = e^t$$

$$0 = 3u^2 - 7u - 6$$

$$0 = (3u+2)(u-3)$$

$$\therefore u = -2/3, 3$$

$$e^t \neq -2/3$$

NO SOL'N

$$e^t = 3$$

$$\frac{\ln 3}{\ln e} = t$$

$$\therefore t = \ln 3 \text{ secs}$$

$$\therefore P(\text{EVEN \# SUM} > 7) = \frac{9}{36}$$

$$(36 \text{ OUTCOMES } 6 \times 6) = 1/4$$

$$\text{ii. } P(\text{AT LEAST 1 EVEN})$$

$$= 1 - P(\text{NOT EVEN})$$

$$= 1 - (3/6 \times 3/6)$$

$$= 3/4$$

OF 36 OUTCOMES

$$3/4 \times 36 = 27 \text{ AT LEAST 1 EVEN.}$$

FROM 9 OUTCOMES IN (i)
6 HAVE AT LEAST 1 EVEN.

$$\therefore 6/27$$

$$\text{bi. } y = e^{-x} + 1$$

ii.

$$\text{iii. } y = e^{-\ln x} + 1$$

$$y = \frac{1}{x} + 1$$

D: All real $x, x > 0$

R: All real $y, y > 1$

$$\text{bi. } \cos \theta = \frac{BD}{a}$$

$$\therefore BD = a \cos \theta$$

$$\sin \theta = \frac{CD}{a}$$

$$\therefore CD = a \sin \theta$$

QUESTION 19.

ai.

<u>8</u>	<u>9</u>	<u>10</u>	<u>11</u>	<u>12</u>
2,6	3,6	4,6	5,6	6,6
3,5	4,5	5,5	6,5	
4,4	5,4	6,4		
5,3	6,3			
6,2				

ii. ADDING UP ALL SIDES...

$$L = 4 + 2a \cos \theta + a \sin \theta + a$$

PTD

$$\text{iii. } L = 4 + 6\sqrt{5} \cos \theta + 3\sqrt{5} \sin \theta + 3\sqrt{5}$$

$$L' = -6\sqrt{5} \sin \theta + 3\sqrt{5} \cos \theta$$

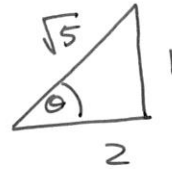
$$\text{STAT. PTS: } 0 = -6\sqrt{5} \sin \theta + 3\sqrt{5} \cos \theta$$

$$6\sqrt{5} \sin \theta = 3\sqrt{5} \cos \theta$$

$$6\sqrt{5} \frac{\sin \theta}{\cos \theta} = 3\sqrt{5}$$

$$\tan \theta = \frac{3\sqrt{5}}{6\sqrt{5}}$$

$$\tan \theta = 1/2 \quad \text{DON'T USE CALC, USE } \Delta:$$



$$\therefore \sin \theta = \frac{1}{\sqrt{5}} \quad \& \quad \cos \theta = \frac{2}{\sqrt{5}}$$

TEST NATURE:

$$L'' = -6\sqrt{5} \cos \theta - 3\sqrt{5} \sin \theta$$

$$= -6\sqrt{5} \times \frac{2}{\sqrt{5}} - 3\sqrt{5} \times \frac{1}{\sqrt{5}}$$

$$= -12 - 3 = -15 \quad -\vee\epsilon \quad \cap \quad \therefore \text{MAX}$$

$\therefore$ MAX LENGTH:

$$L = 4 + \left(6\sqrt{5} \times \frac{2}{\sqrt{5}} \right) + \left(3\sqrt{5} \times \frac{1}{\sqrt{5}} \right) + 3\sqrt{5}$$

$$= 4 + 12 + 3 + 3\sqrt{5}$$

$$= 19 + 3\sqrt{5} \text{ cm.} \quad \text{DONE!}$$